Assignment 2

AI in My City

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Assignment Exercises:

Complete the following exercises in order. For each exercise, show your code, the output, and a brief explanation of what the result means.

Part 1 – Search: Extend the City Map

The map already has 6 cities: Makkah, Jeddah, Taif, Riyadh, Madinah, Dammam. You will add 2 new cities and search for a route to one of them.

Run the File:

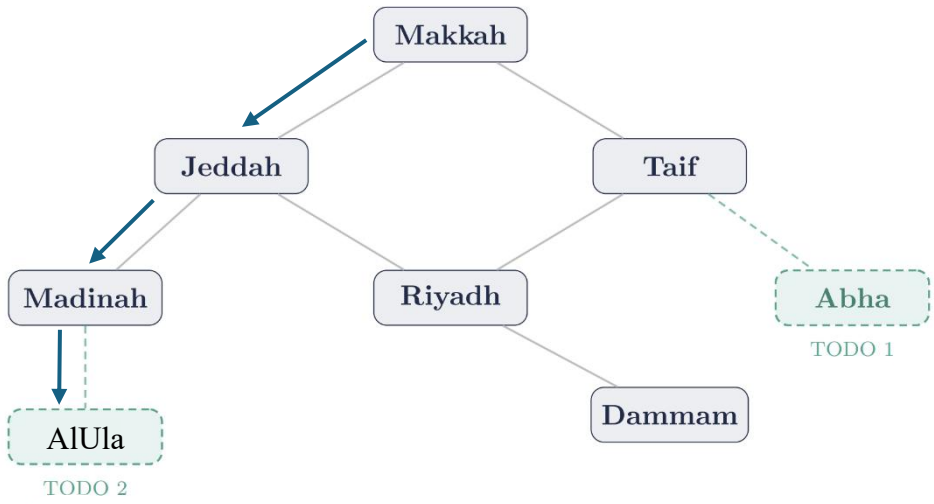
After completing all 4 TODOs, open the terminal in VS Code and run:

```
Searching from 'Makkah' to 'AlUla' using BFS...

 Step 1: Visiting 'Makkah'
 Step 2: Visiting 'Jeddah'
 Step 3: Visiting 'Taif'
 Step 4: Visiting 'Riyadh'
 Step 5: Visiting 'Madinah'
 Step 6: Visiting 'Abha'
 Step 7: Visiting 'Dammam'
 Step 8: Visiting 'AlUla'

Goal reached! 'AlUla' found.

Path found (3 steps):
```



Part 1 Answer Sheet

Fill in your answers below after running the program.

Answer 1 – Your New City

The name of the city I added (TODO 2):

City name: AlUla

It is connected to: Madinah

Answer 2 – BFS Result

Fill in the blanks using what you saw in the terminal output.

a. BFS started at: Makkah

b. BFS was looking for: AlUla

c. The path BFS found was:

Makkah → Jeddah → Madinah → AlUla

(add more arrows if your path is longer)

d. The path had 3 steps (hops) in total.

e. Did BFS visit any cities that were NOT on the final path? **Yes** ■ **No** □

If yes, name one: Riyadh

f. Why does BFS visit extra cities that are not on the final path?

Because BFS explores all neighbors at the current depth level (layer by layer).

Part 2 – Knowledge: Complete the Logic Detective

Your Task

Open assign2_part2_knowledge.py.

The Knowledge Base already has 4 clues and 5 symbols. You need to:

• Add **1 missing symbol** (nora_suspicious)

• Add **2 missing clues** (Clues 5 and 6) to the KB

Run the File

After completing all 4 TODOs, run:

```
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  CS3081 Assignment 2 - Knowledge Base Detective
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 Query 1 : Is Lina guilty?
   Answer: YES - KB entails Lina is guilty.

 Query 2 : Is Nora suspicious?
   Answer: NO  - KB does not entail Nora is suspicious.

 Query 3 : Does Sara have a key?
   Answer: NO  - KB does not entail Sara has a key.

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```

Part 2 Answer Sheet

Answer 3 – Query Results

Write the answer your program printed for each query.

a. Query 1 – Is Lina guilty?
   Program answered: **YES** ■ **NO** □

b. Query 2 – Is Nora suspicious?
   Program answered: **YES** □ **NO** ■

c. Query 3 – Does Sara have a key?
   Program answered: **YES** □ **NO** ■

Answer 4 – Understanding the Results

a. Why is Lina guilty according to the KB? (Trace the clues: which clues lead to this conclusion?)

Because in the clues it says (If Lina was seen → Lina has a key) and (Lina was seen near the lab) and (If Lina has a key → Lina is guilty).

b. Clue 5 says Nora HAS an alibi. Yet Query 2 asks if she is suspicious. Based on the program's answer – is she suspicious?

No, she isn't.

In your own words, why?

Because in clue 6 there is a condition said kb.tell(Implication(Not(nora_alibi), nora_suspicious)).

c. Complete this sentence in your own words:

"The Knowledge Base cannot prove a statement unless . . ."

all the statements are True in every model where the KB is True.

Reflection Question

This question connects both parts of the assignment. Answer in 2–4 sentences.

Reflection – Search vs. Knowledge

In Part 1, BFS ***searched*** *for a path. In Part 2, the KB* ***deduced*** *a fact.*

What is the difference between what BFS does and what the KB does?
Think about: What is the input? What is the output? What kind of problem does each solve?

The BFS is a search algorithm which use (FIFO) and it search level by level and it guarantee to find the shortest path. While KB is a knowledge base where it comes True when all the statement is True, then KB entail α.